5.8 A Hidden Distance in the Disc

WORKSHEET

The question

Let z and w be complex numbers. Use a diagram to show that $|z-w| \leq |z|+|w|$.

For any complex numbers z and w, E is defined by

$$E = zw^* + z^*w + 2|zw|.$$

(i) Show that $|z-w|^2 = \left(|z|+|w|\right)^2 - E$, and deduce that E is real and non-negative.

(ii) Show that $|1-zw^*|^2 = \left(1+|zw|\right)^2 - E$.

Hence show that, if both $|z|>1$ and $|w|>1$, then

$$\frac{|z-w|}{|1-zw^*|} \leq \frac{|z|+|w|}{1+|zw|}.$$

Does this inequality also hold if both $|z|<1$ and $|w|<1$?

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself How does $|z-w| \le |z|+|w|$ follow from a diagram? (Think of z and $-w$ as vectors.)

STAGE 1 Expanding the moduli

Ask yourself Expand $|z-w|^2 = (z-w)(z^*-w^*)$. Compare with $(|z|+|w|)^2$.

STAGE 3 The inequality

Ask yourself Write $A = (|z|+|w|)^2$ and $B = (1+|zw|)^2$. The inequality, squared, says $\frac{A-E}{B-E} \le \frac{A}{B}$. When is that true?

Pause and try Show that $1+|z||w|-|z|-|w|$ factorises. When is it non-negative?

Try this next

1. **A numerical check.** Take $z = \frac{1}{2}$ and $w = -\frac{1}{2}$. Compute both sides of the inequality. Then compute the hyperbolic distances $d(z,0)$, $d(0,w)$ and $d(z,w)$ and check the triangle inequality is an equality here. Why?

2. **The mixed case.** Find z, w with $|z| < 1 < |w|$ for which the inequality fails.

3. **An isometry.** Let $f(z) = \dfrac{z-a}{1-a^*z}$ with $|a| < 1$. Show that $|f(z)| = 1$ whenever $|z| = 1$, and that $f(a) = 0$. Show that $\rho(f(z), f(w)) = \rho(z,w)$.

4. **Infinity is near.** Show that the hyperbolic distance from 0 to r (with $0 < r < 1$) is $\ln \frac{1+r}{1-r}$, and that it tends to infinity as $r \to 1$. How close to the boundary must a point be to be hyperbolic distance 10 from the centre?

Further reading

- **Caroline Series, "Hyperbolic geometry" lecture notes** (University of Warwick, freely available). A clear first course, starting from the Poincaré disc.
- **Jeremy Gray, *Ideas of Space*** (Oxford University Press). The history of non-Euclidean geometry from Euclid to Einstein.

- **Tristan Needham, *Visual Complex Analysis*** (Oxford University Press). Chapter 6 covers non-Euclidean geometry and Möbius transformations, with beautiful pictures.

The full guided solution, and what the question leads to at university, is in Section 5.8:
https://beyondstep.keithmcnulty.org/ch05/poincare-disc.html